

Original Article

Maternal health in pregnancy and associations with adverse birth outcomes: Evidence from Growing Up in New Zealand

Amy L. BIRD,^{1,2} Cameron C. GRANT,^{1,3,4,5} Dinusha K. BANDARA,¹ Jatender MOHAL,¹ Polly E. ATATOCA-CARR,^{1,3,6} Michelle R. WISE,⁷ Hazel INSKIP,⁸ Motohide MIYAHARA⁹ and Susan M.B. MORTON^{1,3}

¹*Growing Up in New Zealand*, ²*Department of General Practice and Primary Health Care*, ³*Centre for Longitudinal Research – He Ara ki Mua*, ⁴*Department of Paediatrics: Child and Youth Health, The University of Auckland*, ⁵*Starship Children's Hospital, Auckland District Health Board, Auckland*, ⁶*National Institute of Demographic and Economic Analysis, University of Waikato, Hamilton*, ⁷*Department of Obstetrics & Gynaecology, University of Auckland, Auckland, New Zealand*, ⁸*MRC Lifecourse Epidemiology Unit, University of Southampton, Southampton, UK* and ⁹*School of Physical Education, Sport and Exercise Sciences, University of Otago, Dunedin, New Zealand*

Objective: To examine prospectively multiple indicators of pregnancy health and associations with adverse birth outcomes within a large, diverse sample of contemporary women.

Design: A cohort of pregnant women who gave birth during 2009–10.

Population: We enrolled a sample of 6822 pregnant New Zealand (NZ) women: 11% of all births in NZ during the recruitment period.

Methods: We analysed a number of maternal health indicators and behaviours during pregnancy in relation to birth outcomes using multivariable logistic regression. Associations were described using adjusted odds ratios and 95% confidence intervals.

Main outcome measures: Three birth outcomes, low birth weight (LBW), pre-term birth (PTB) and delivery type, were measured via linkage with maternity hospital perinatal databases. Small for gestational age (SGA) was then defined as below the 10th percentile by week of gestation.

Results: Modelling of birth outcomes after adjusting for confounders indicated patterns of increased risk of LBW and PTB for women who smoke, have elevated pre-pregnancy body mass index (BMI), or with insufficient pregnancy weight gain. SGA was associated with maternal smoking, alcohol use, insufficient weight gain and nausea and vomiting during pregnancy. Risk of caesarean section was associated with having a diagnosed illness before pregnancy, elevated BMI, greater pregnancy weight gain and less pregnancy exercise. Number of risk factor variables were then used to model birth outcomes. Women with multiple risk factors were at increased risk compared with those who had no risk factors.

Conclusions: Women with multiple health risks are at particular risk of adverse birth outcomes.

Key words: antenatal health, caesarean, low birth weight, pre-term birth, small for gestational age.

Introduction

Pregnancy is a significant life event with enormous potential to impact a woman's health, and the subsequent health and wellbeing of her child. Previously well women may suffer

pregnancy-related health issues, or conversely, may have greater contact with health services and make deliberate efforts to manage diet and weight, smoking and alcohol use.¹

Several decades of research has identified maternal smoking and alcohol use, maternal weight and illness or disability as key correlates of adverse birth outcomes.^{2–5} Physical activity does not appear to increase the risk of low birth weight (LBW) or pre-term birth (PTB)^{6,7} and may reduce the likelihood of caesarean delivery.⁸ These critical findings have contributed to widespread interventions, public health campaigns and increased public knowledge.

The current study will address two key issues in the literature. First, we need up-to-date information from contemporary, diverse, prospective cohorts that examine

Correspondence: Dr Amy L. Bird, Growing Up in New Zealand, The University of Auckland Tamaki Campus, 261 Morrin Road, Glen Innes, Auckland 1072, New Zealand. Email: a.bird@auckland.ac.nz

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associations between maternal pregnancy health and birth outcomes. On the one hand increasing awareness about pregnancy health may have lessened the population impact of maternal health behaviours on birth outcomes. On the other hand, some health risk factors are increasing in the general population (e.g., obesity⁹) and others are changing differentially (e.g., smoking is typically declining among Western higher socio-demographic groups but continues to increase for some ethnic minority populations¹⁰).

Second, much of our knowledge comes from studies that focus on one or two indicators of health (e.g., smoking or alcohol; disability or specific diseases), yet in reality women present clinically with multiple health indicators, and risk behaviours during pregnancy are likely to cluster. For example, women who smoke during pregnancy are also more likely to use alcohol¹¹ and to have a disability.¹² Researchers have adopted a cumulative impact approach to understanding the multiple risk factors that contribute to childhood obesity.^{13,14} The application of this approach to understanding pregnancy outcomes would have direct clinical and public health value, particularly for New Zealand (NZ) women. Contemporary NZ is unique in its socio-demographic diversity, as well as ethnic health inequalities. Our alcohol and smoking rates are high, particularly among young women, and our obesity rate is the third highest in the Organisation for Economic Co-operation and Development.¹⁵ There is some evidence that Maori and Pacific women, and women who smoke, may be more likely to drink alcohol during pregnancy.¹⁶ Understanding the cumulative impact of health risk factors during pregnancy for the birth outcomes of NZ women will have direct relevance for clinicians working with, and public health interventions tailored for, our most vulnerable women.^{13,14}

Our aim was to examine associations of pregnancy health with the subsequent risk of LBW, PTB, small for gestational age (SGA) and caesarean delivery among a contemporary, diverse population cohort. We were particularly interested in the risk of adverse birth outcomes for women who experience multiple health risk factors. To achieve this we utilised a large, diverse cohort of NZ women recruited during pregnancy.

Materials and Methods

Participants

Our study was completed within NZ's child cohort, Growing Up in New Zealand (www.growingup.co.nz). Ethics approval was granted by the Northern Y Regional Ethics Committee for data collection with participants and data linkage (NTY/08/06/055). All participants provided written, informed consent. Antenatal recruitment, engagement with an ethnically and socioeconomically diverse sample and inclusion of partners were essential study design features.^{17,18} The recruited cohort of 6822 pregnant women resided within a geographically defined region, where 29% of NZ's population lives. We enrolled 11% of the national birth cohort born during the study recruitment

period (April 2009 to March 2010). Our enrolled cohort of 6853 children is aligned to all NZ births.¹⁹

Data collection

We completed a computer-assisted face-to-face interview with each enrolled woman in the last trimester of pregnancy. Birth outcomes were obtained by data linkage with perinatal medical records. Consent to link to these records was obtained for 6682 (98%) of the 6853 enrolled infants.

Measurements

Doctor-diagnosed illness

We asked women if they had doctor-diagnosed diabetes, asthma, heart disease, high blood pressure or anaemia, and when. Responses were recoded to reflect any of these doctor-diagnosed illnesses before pregnancy, both before and during, or only during this pregnancy.

Disability

Women were asked whether they had a disability lasting more than 6 months²⁰ before this pregnancy, during or both before and during pregnancy.

Body mass index (BMI) and weight change

We asked each woman to self-report her height, pre-pregnancy weight and current weight. Both self-reported pre-pregnancy weight and height have been validated against independent measures of BMI.²¹ Weight change was calculated by subtracting pre-pregnancy weight from current weight at the time of interview.

Cigarette smoking

We asked each woman to estimate the average number of cigarettes smoked per day before and during the pregnancy. Responses were grouped into: those who did not smoke (both before and during pregnancy); those who stopped smoking for pregnancy; or those who continued to smoke during pregnancy.

Alcohol use

We asked each woman to estimate the average number of alcoholic drinks consumed per week before pregnancy, during the first three months of pregnancy, and during the remainder of pregnancy. Questions were derived from NZ's National Nutrition Survey.²² Responses were further categorised into no alcohol, less than one drink per week, one drink per week, or more than two drinks per week.

Activity

Each woman was asked to estimate the frequency (days per week) and duration (<30 min, 30–60 min, >60 min)

of both vigorous and moderate activity: before pregnancy, during the first trimester of pregnancy and for the remainder of pregnancy.²³ Women who engaged in moderate activity for at least 30 min on at least five out of seven days, or who engaged in vigorous activity for at least 30 min at least two out of seven days were classified as engaging in 'regular' activity.

Birth outcomes

Women gave consent for linkage to the birth information collected by maternity hospitals and District Health Boards. For multiple pregnancies birth outcomes were classified for the first infant. LBW was classified as <2500 g, PTB as <37 weeks, and delivery mode as caesarean versus all other delivery modes. SGA was calculated using the World Health Organisation calculator where infants with birth weights <10th percentile for each week of gestation were classified as SGA.

Socio-demographic measures

Women were asked a range of standard questions at the antenatal interview. Area-level socio-economic deprivation was measured using the NZ Index of Deprivation.²⁴ Maternal education was categorised based on highest qualifications. Women were asked to self-prioritise their ethnicity and responses were grouped into European, Maori, Pacific, Asian and Other categories. Labour force participation was categorised as employed, unemployed, student or not in workforce.

Data analysis

The cohort was specifically designed to have adequate power to undertake complex analyses within ethnic and socioeconomic subgroups as well as for the whole cohort.¹⁷ We used descriptive statistics to report measures of health. We developed logistic regression models to examine associations of these maternal health measures with the risk of each of the adverse birth outcomes: LBW, PTB, SGA and caesarean section delivery. We reported independent associations using adjusted odds ratios (OR) and 95% confidence intervals (CI). Analyses were conducted using SAS software (version 9.4, SAS Institute Inc., Cary, NC, US). A two-sided *P*-value of <0.05 was considered statistically significant.

Results

Descriptive analyses

Descriptive statistics for pregnancy health measures are shown in Table 1. Over 40% of women reported weight that fell in the overweight or obese range. Of those who smoked before pregnancy, almost half stopped. Alcohol use also reduced. The proportion of women who engaged in regular exercise also declined.

Associations of maternal health and health behaviours during pregnancy with birth outcomes

Logistic regression models were created where each pregnancy health variable predicted the likelihood of LBW, PTB, SGA or caesarean (see Tables 2 and 3). In

Table 1 Maternal health before and during pregnancy

Maternal health measure	<i>n</i> (%)
Doctor-diagnosed illness	
Before pregnancy only† (<i>n</i> = 6822)	1994 (29)
Before and during pregnancy‡ (<i>n</i> = 6822)	1199 (18)
Pregnancy only§ (<i>n</i> = 6822)	615 (9)
Current disability (<i>n</i> = 6179)	
Yes¶	384 (6)
No	5795 (94)
Maternal pre-pregnancy BMI (<i>n</i> = 5971)	
Underweight	256 (4)
Normal	3261 (55)
Overweight	1349 (23)
Obese	1105 (19)
Weight gain during pregnancy (<i>n</i> = 5507)	
0 (or weight loss)	223 (4)
1–5 kg	672 (12)
6–10 kg	2079 (38)
11–15 kg	1588 (29)
16–20 kg	597 (11)
>20 kg	348 (6)
Maternal smoking (<i>n</i> = 6161)	
No smoking (before or during)	4904 (80)
Stopped smoking for pregnancy	610 (10)
Continued smoking during pregnancy	647 (10)
Alcohol use before pregnancy (<i>n</i> = 6805)	
No alcohol	1982 (29)
<1 drink per week	1238 (18)
1 drink per week	432 (6)
2+ drinks per week	3153 (46)
Alcohol use first trimester (<i>n</i> = 6804)	
No alcohol	5266 (77)
<1 drink per week	613 (9)
1 drink per week	168 (2)
2+ drinks per week	757 (11)
Alcohol use rest of pregnancy (<i>n</i> = 6810)	
No alcohol	5898 (87)
<1 drink per week	635 (9)
1 drink per week	112 (2)
2+ drinks per week	165 (2)

†1007 (15%) had asthma, 335 (5%) had heart disease or high blood pressure, 944 (14%) had anaemia, 49 (1%) had diabetes.

‡551 (8%) had asthma, 127 (2%) had heart disease or high blood pressure, 559 (8%) had anaemia, 80 (1%) had diabetes.

§24 (0.03%) had asthma, 154 (2%) had heart disease or high blood pressure, 317 (5%) had anaemia, 162 (2%) had diabetes.

¶134 (2.0%) of the sample had a sensory disability, 114 (1.7%) had a mobility/agility disability, 83 (1.2%) had a psychological or intellectual disability, 95 (1.4%) indicated a disability in some other area of functioning.

order to adjust for the potential effect of socio-demographic and other pregnancy-related variables, each model also included maternal age, ethnicity, area deprivation, maternal education, labour force participation, parity and pregnancy planning. Women with a doctor-diagnosed illness before (OR = 1.33), but not during (OR = 1.21), pregnancy had a slightly increased risk of caesarean delivery. Women with a doctor-diagnosed illness during (OR = 1.74), but not before (OR = 0.85), pregnancy were more likely to have a LBW infant.

Women were more likely to have a LBW infant if they had a pre-pregnancy BMI in the overweight (OR = 1.42) or obese (OR = 1.65) range. Women were also more likely to have a PTB infant if their pre-pregnancy BMI was in the overweight (OR = 1.45) or obese (OR = 1.61) range. Women were more likely to have a caesarean delivery if their pre-pregnancy BMI was in the overweight (OR = 1.46) or obese range (OR = 1.81).

Women who lost weight during pregnancy were more likely than those who gained a moderate amount of weight (6–10 kg) to have a LBW infant (OR = 2.03) and a PTB (OR = 1.91). Women who lost weight (OR = 1.78) or only gained a small (1–5 kg) of weight (OR = 1.38) were more likely to have a SGA infant. Women who gained 16–20 kg were less likely to have a LBW infant (OR = 0.41). Women who gained 11–15 kg were less likely to have a PTB (OR = 0.70). Women who gained 11–15 kg (OR = 0.77) or 16–20 kg (OR = 0.45) were less likely to have a SGA infant. Women were more likely to have a caesarean delivery if they gained 11–15 kg (OR = 1.20), 16–20 kg (OR = 1.25) or more than 20 kg (OR = 1.55).

Compared with non-smokers, women who continued to smoke during pregnancy were more likely to have a LBW infant (OR = 1.93) and PTB (OR = 1.63). In contrast, women who stopped smoking were not at significantly increased risk of having a LBW infant (OR = 1.47) or

Table 2 Adjusted logistic regressions: maternal health status, smoking and alcohol modelling birth outcomes

	LBW: OR (95% CI)†	PTB: OR (95% CI)†	SGA: OR (95% CI)†	Caesarean: OR (95% CI)†
Doctor-diagnosed illness before pregnancy				
No	1.00	1.00	1.00	1.00
Yes	0.85 (0.63–1.13)	1.02 (0.80–1.29)	1.04 (0.86, 1.26)	1.33 (1.17–1.52)****
Doctor-diagnosed illness before and during pregnancy				
No	1.00	1.00	1.00	1.00
Yes	1.00 (0.71–1.37)	1.12 (0.85–1.46)	1.01 (0.81, 1.26)	1.08 (0.92–1.25)
Doctor-diagnosed illness during pregnancy				
No	1.00	1.00	1.00	1.00
Yes	1.74 (1.20–2.47)**	1.38 (0.97–1.92)	1.28 (0.96, 1.67)	1.21 (0.99–1.48)
Current disability				
No	1.00	1.00	1.00	1.00
Yes	1.26 (0.76–1.98)	0.95 (0.59–1.47)	1.19 (0.83, 1.65)	1.25 (0.98–1.58)
Smoking				
Any smoking during pregnancy	1.93 (1.24–2.95)*	1.63 (1.10–2.36)*	2.16 (1.60, 2.90)****	0.76 (0.58–0.98)*
Stopped smoking	1.47 (0.93–2.25)	1.02 (0.66–1.51)	1.47 (1.09, 1.96)*	1.15 (0.93–1.42)
Non-smokers	1.00	1.00	1.00	1.00
Alcohol use before pregnancy				
No alcohol	1.00	1.00	1.00	1.00
<1 drink per week	0.82 (0.56–1.19)	0.90 (0.65–1.23)	0.82 (0.63, 1.07)	0.99 (0.82–1.18)
1 drink per week	0.88 (0.49–1.51)	0.94 (0.58–1.48)	0.93 (0.62, 1.36)	1.08 (0.83–1.39)
2+ drinks per week	0.78 (0.56–1.08)	0.75 (0.57–1.00)	1.01 (0.81, 1.26)	0.91 (0.77–1.06)
Alcohol use in first trimester				
No alcohol	1.00	1.00	1.00	1.00
<1 drink per week	0.85 (0.52–1.31)	1.06 (0.73–1.51)	1.20 (0.89, 1.59)	0.88 (0.71–1.08)
1 drink per week	0.95 (0.40–1.92)	0.74 (0.31–1.49)	1.04 (0.56, 1.77)	0.68 (0.43–1.02)
2+ drinks per week	0.70 (0.43–1.08)	0.78 (0.52–1.13)	1.28 (0.97, 1.66)	0.98 (0.80–1.20)
Alcohol use in rest of pregnancy				
No alcohol	1.00	1.00	1.00	1.00
<1 drink per week	0.70 (0.42–1.11)	0.71 (0.46–1.05)	1.24 (0.92, 1.65)	0.79 (0.64–0.97)
1 drink per week	1.05 (0.36–2.38)	0.44 (0.11–1.18)	1.46 (0.75, 2.61)	0.84 (0.52–1.33)
2+ drinks per week	0.54 (0.16–1.30)	0.49 (0.17–1.10)	1.81 (1.11, 2.84)*	1.16 (0.78–1.69)

* $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$.

†Each logistic regression modelled a pregnancy health variable on a birth outcome, and all were adjusted for maternal age, ethnicity, area deprivation, maternal education, labour force participation, parity and pregnancy planning.

LBW, low birth weight; PTB, pre-term birth; SGA, small for gestational age

Table 3 Adjusted logistic regressions: maternal weight, activity, and nausea modelling birth outcomes

	LBW: OR (95% CI)†	PTB: OR (95% CI)†	SGA: OR (95% CI)†	Caesarean: OR (95% CI)†
Pre-pregnancy BMI				
<18.5 Underweight	1.18 (0.60–2.11)	1.15 (0.61–2.00)	1.23 (0.82, 1.79)	0.67 (0.45–0.95)*
18.5 ≥ Normal < 25	1.00	1.00	1.00	1.00
25 ≥ Overweight < 30	1.42 (1.02–1.95)*	1.45 (1.10–1.92)*	1.04 (0.83, 1.30)	1.46 (1.24–1.70)****
≥30 Obese	1.65 (1.13–2.37)*	1.61 (1.17–2.20)*	0.89 (0.67, 1.17)	1.81 (1.51–2.17)****
Weight change during pregnancy				
0 or weight loss	2.03 (1.14–3.42)*	1.91 (1.14–3.05)**	1.78 (1.16, 2.65)**	1.20 (0.85–1.69)
1–5 kg	1.26 (0.84–1.86)	1.31 (0.92–1.85)	1.38 (1.05, 1.80)*	0.88 (0.71–1.10)
6–10 kg	1.00	1.00	1.00	1.00
11–15 kg	0.79 (0.56–1.12)	0.70 (0.51–0.95)*	0.77 (0.61, 0.97)*	1.20 (1.03–1.40)*
16–20 kg	0.41 (0.20–0.76)**	0.70 (0.43–1.08)	0.45 (0.30, 0.66)****	1.25 (1.00–1.55)*
>20 kg	1.22 (0.67–2.11)	1.40 (0.86–2.20)	0.83 (0.54, 1.24)	1.55 (1.16–2.04)*
Pre-pregnancy activity: ≥5 moderate or ≥2 vigorous				
No	1.00	1.00	1.00	1.00
Yes	0.93 (0.72, 1.21)	0.99 (0.79, 1.24)	1.09 (0.91, 1.30)	0.92 (0.81, 1.04)
First trimester activity: ≥5 moderate or ≥2 vigorous				
No	1.00	1.00	1.00	1.00
Yes	0.97 (0.73, 1.29)	0.98 (0.76, 1.24)	1.14 (0.95, 1.37)	0.89 (0.78, 1.02)
Rest of pregnancy activity: ≥5 moderate or ≥2 vigorous				
No	1.00	1.00	1.00	1.00
Yes	0.82 (0.59, 1.11)	0.81 (0.61, 1.06)	1.08 (0.89, 1.32)	0.74 (0.64, 0.86)****
First trimester nausea				
None	1.00	1.00	1.00	1.00
Mild	0.82 (0.58, 1.15)	0.84 (0.62, 1.15)	1.04 (0.83, 1.32)	0.97 (0.82, 1.15)
Moderate	0.89 (0.62, 1.26)	1.15 (0.85, 1.57)	0.89 (0.69, 1.14)	1.11 (0.94, 1.32)
Severe	0.96 (0.67, 1.38)	1.10 (0.80, 1.52)	1.32 (1.04, 1.68)*	1.05 (0.87, 1.26)
Rest of pregnancy nausea				
None	1.00	1.00	1.00	1.00
Mild	1.29 (0.94, 1.74)	1.26 (0.96, 1.63)	1.35 (1.09, 1.66)**	1.14 (0.98, 1.33)
Moderate	1.09 (0.71, 1.61)	0.95 (0.65, 1.35)	1.11 (0.83, 1.45)	1.19 (0.98, 1.43)
Severe	1.07 (0.57, 1.85)	1.28 (0.77, 2.01)	1.91 (1.35, 2.65)***	0.98 (0.73, 1.30)

* $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$.

†Logistic regressions adjusted for maternal age, ethnicity, area deprivation, maternal education, labour force participation, parity and pregnancy planning.

BMI, body mass index; LBW, low birth weight; PTB, pre-term birth; SGA, small for gestational age

PTB (OR = 1.02) compared with non-smokers. Women who continued to smoke were less likely than non-smokers to have a caesarean delivery (OR = 0.76). Both women who continued to smoke (OR = 2.16), and women who were previously smokers but stopped for pregnancy (OR = 1.47) were more likely than non-smokers to have a SGA infant.

Women who engaged in regular exercise (at least two sessions of vigorous activity or five sessions of moderate activity per week) during the second and third trimesters of pregnancy were less likely to have a caesarean delivery (OR = 0.74).

Women who experienced regular severe nausea and vomiting during the first trimester of pregnancy were more likely to have a SGA infant (OR = 1.32). Women who experienced mild (OR = 1.35) or severe (OR = 1.91) nausea during the rest of the pregnancy were more likely to have a SGA infant.

Multiple health risk factors and adverse birth outcomes

Based on the factors previously identified as significant predictors of adverse birth outcomes, variables were calculated that described the number of risk factors present.^{18,19} The highest risk factor categories for LBW and PTB were combined due to low frequencies. Logistic regression models were then created with number of risk factors predicting risk of adverse birth outcome, adjusting for socio-demographic confounders (Table 4). Women with one (OR = 1.51), two (OR = 2.24) or three or four (OR = 3.86) risk factors were at increasing risk of having a LBW infant. Women with two or three risk factors (OR = 2.87) were at greater risk of PTB. Women with one (OR = 1.72), two (OR = 2.39) or three (OR = 6.24) risk factors were at increasing risk of having a SGA infant. Women with two (OR = 2.03), three (OR = 2.86) or four

Table 4 Adjusted logistic regressions: number of health risk factors modelling birth outcomes

Number of risk factors	Risk of birth outcome					
	LBW: OR (95% CI) [†]	Number of risk factors	PTB: OR (95% CI) [†]	Number of risk factors	SGA: OR (95% CI) [†]	Number of risk factors
0	1.00	0	1.00	0	1.00	0
1	1.51 (1.09, 2.08)*	1	1.25 (0.94, 1.65)	1	1.72 (1.38, 2.15) ****	1
2	2.24 (1.40, 3.52)***	2 or 3	2.87 (1.89, 4.28)****	2	2.39 (1.60, 3.50) ****	2
3 or 4	3.86 (1.66, 8.11)***			3	6.24 (1.30, 23.24)**	3
						4
Risk factors						
- BMI overweight or obese - Weight loss/no weight gain (pregnancy) - Continuing to smoke - Doctor diagnosed illness (pregnancy)						
- BMI overweight or obese - Weight loss/no weight gain (pregnancy) - Continuing to smoke						
- Weight loss/no weight gain (pregnancy) - Smoking (either before or during pregnancy) - More than 2 + alcoholic drinks (2nd and 3rd trimesters) - Pregnancy nausea (severe 1st trimester or mild or severe rest of pregnancy)						
- BMI overweight or obese - Weight gain >11 kg (pregnancy) - Doctor diagnosed illness (before pregnancy) - Lack of regular exercise (2nd and 3rd trimesters)						

* $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$.[†]Logistic regressions adjusted for maternal age, ethnicity, area deprivation, maternal education, labour force participation, parity, and pregnancy planning. BMI, body mass index; LBW, low birth weight; PTB, pre-term birth; SGA, small for gestational age

(OR = 4.56) risk factors were at increased risk of caesarean delivery.

Discussion

Main findings

In this prospective study of women's pregnancy health we found multiple independent associations between health and behaviours, and subsequent birth outcomes after adjusting for a range of socio-demographic and pregnancy-related cofounders. We also found that having multiple health risk factors further increased women's risk of adverse birth outcomes.

New Zealand and international guidelines caution against any maternal smoking or alcohol use during pregnancy.^{25,26} While almost half of women who smoked stopped for pregnancy, and overall alcohol use dropped, continuing to smoke and higher alcohol consumption were associated with increased risks of adverse birth outcomes. Despite zero-tolerance policy and public health messages, continued efforts are needed to support pregnant women. The rewards are clear: with the exception of SGA, ceasing smoking and alcohol once pregnant eliminated increased risks for adverse birth outcomes.

Maternal weight emerged as a strong predictor of adverse birth outcomes. Interestingly, pre-pregnancy BMI and weight gain during pregnancy showed different patterns. While weight loss during pregnancy was associated with increased risk of LBW, PTB and SGA, it was maternal pre-pregnancy overweight or obesity, not underweight, that predicted increased risks of PTB, LBW and delivery by caesarean. This emphasises the importance for clinicians of considering both pre-pregnancy BMI and weight gain during pregnancy, and the need for effective population health and policy approaches that target healthy weight management for women of child-bearing age.²⁷ Regular, sustained physical activity is one way of managing pregnancy weight gain, and current recommendations generally encourage women to continue with regular moderate exercise.⁶ Our findings support previous research in this area (typically limited by small samples): pregnancy exercise was not associated with increased risk of adverse birth outcomes, and was in fact associated with reduced risk for caesarean delivery.⁸

Pregnancy nausea and vomiting is often viewed as protective against adverse birth outcomes,²⁸ yet the current findings indicate that women with severe nausea and vomiting throughout pregnancy, and mild nausea during the latter parts of pregnancy were at increased risk of having a SGA infant. It is possible that these findings may be explained by the presence of comorbid depression and/or the more severe presentation of hyperemesis gravidum. Nevertheless, the findings here highlight the importance of clinical assessment and management of persistent nausea and vomiting during pregnancy.

Women with multiple health risk factors were at increased risk of adverse birth outcomes, and risk

increased with each added risk factor. These findings highlight the importance of identifying and supporting women who present with more than one health risk factor (Fig. 1).

Strengths and limitations

The prospective, longitudinal design of the current study reduces recall bias. The large, ethnically diverse sample enhances the generalisability of the findings. A further strength is the measurement of multiple measures of health and health behaviours within a population-based sample.

One limitation of this study is that measures of health and behaviour during pregnancy were based on self-reports. In terms of BMI, self-reporting and independent measures of height and weight are generally highly correlated, although heavier individuals and women are more likely to underestimate their weight, and shorter individuals more likely to overestimate their height.²¹ Despite this, BMI calculations based on self-reports are widely used, particularly in very large samples where independent measurement is simply not practical.

The inclusion of caesarean delivery as a birth outcome warrants further discussion. Caesarean rates are increasing internationally and vary widely across countries. A population caesarean rate between 10 and 15% is considered optimal for both mother and child safety and economic impact, yet most countries – including New Zealand – sit outside this range.²⁹ Caesarean rates are influenced by a range of socio-demographic, clinician and public policy factors, thus associations with health are more likely to be influenced by potential confounding variables than those for LBW, PTB and SGA.

Causality cannot be inferred from these results, and although a range of socio-demographic and maternal variables were included in multivariable analyses, potentially, other confounding variables may underlie associations.³⁰ In addition, birth outcomes are an important clinical indicator, but are by no means the only

- Women who continue to **smoke** and/or consume **alcohol** during pregnancy
- Women with a **doctor-diagnosed illness** (before or during pregnancy)
- Women with a pre-pregnancy BMI in the **overweight or obese** range
- Women who **fail to gain weight** during pregnancy
- Women with severe and/or persistent **nausea and vomiting** during pregnancy

Women with >2 of these risk factors are at 2–6 times the risk of adverse birth outcomes

Figure 1 Overview: women at increased risk of adverse birth outcomes.

important child outcome. For example, maternal alcohol use is more typically associated with neuro-behavioural developmental differences which emerge as children develop. This highlights the importance of considering the current findings as a specific research question within the context of a broader longitudinal study and context.

Conclusion

Despite increasing public health messages, known health risk factors such as weight, smoking, alcohol use and maternal illness continue to be prevalent among a representative sample of pregnant NZ women, and are associated with increased risks of adverse birth outcomes. Women at greatest risk were those with multiple indicators of poor health. Stopping smoking, limiting alcohol use, managing weight gain and engaging in physical exercise were all associated with lower risks of adverse birth outcomes. For women with poorer pre-pregnancy health, or those with unplanned pregnancies, the current findings highlight that behavioural change, even later in pregnancy, is worthwhile and important.

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Contributors' Statements

Amy Bird analysed and interpreted the data and completed the first and final drafts of the manuscript. Cameron Grant contributed to the conception and design of the study, developed the data collection instruments, analysed and interpreted the data and completed the first and final drafts of the manuscript. Dinusha Bandara assisted with the data analysis approach and interpretation of the data, revised the manuscript and approved the final manuscript as submitted. Jatender Mohal assisted with the data analysis approach, revised the manuscript and approved the final manuscript as submitted. Polly Atatoa Carr contributed to the conception and design of the study, revised the manuscript and approved the final manuscript as submitted. Michelle Wise assisted with the data analysis approach, revised the manuscript and approved the final manuscript as submitted. Hazel Inskip contributed to the conception and design of the study and the development of the data collection instruments, assisted with the data analysis approach, revised the manuscript and approved the final manuscript as submitted. Motohide Miyahara contributed to the development of the data collection instruments, revised the manuscript and approved the final manuscript as submitted. Susan Morton conceived and designed the study, developed the data collection instruments, revised the manuscript and approved the final manuscript as submitted.

References

- 1 Crozier SR, Robinson SM, Borland SE *et al.* Do women change their health behaviours in pregnancy? Findings from the Southampton women's survey. *Paediatr Perinat Epidemiol* 2009; **23**: 446–453.
- 2 Shah NR, Bracken MB. A systematic review and meta-analysis of prospective studies on the association between maternal cigarette smoking and preterm delivery. *Am J Obstet Gynecol* 2000; **182** (2): 465–472.
- 3 Salihu HM, Wilson RE. Epidemiology of prenatal smoking and perinatal outcomes. *Early Human Dev* 2007; **83**: 713–720.
- 4 Patra J, Bakker R, Irving H *et al.* Dose–response relationship between alcohol consumption before and during pregnancy and the risks of low birthweight, preterm birth and small for gestational age (SGA)—a systematic review and meta-analyses. *Br J Obstet Gynaecol* 2011; **118**: 1411–1421.
- 5 McDonald SD, Han Z, Mulla S, Beyene J. Overweight and obesity in mothers and risk of preterm birth and low birth weight infants: systematic review and meta-analyses. *Br Med J* 2010; **341**: 1–20.
- 6 Nascimento SL, Surita FG, Cecatti JG. Physical exercise during pregnancy: a systematic review. *Curr Opin Obstet Gynecol* 2012; **24**: 387–394.
- 7 Haakstad LAH, Bo K. Exercise in pregnant women and birth weight: a randomized controlled trial. *BMC Pregnancy Childbirth* 2011; **11**: 66.
- 8 Price BB, Amini SB, Kappeler K. Exercise in pregnancy: effect on fitness and obstetric outcomes – a randomized trial. *Med Sci Sports Exerc* 2012; **44**: 2263–2269.

- 9 Malik VS, Willett WC, Hu FB. Global obesity: trends, risk factors and policy implications. *Nat Rev Endocrinol* 2013; **9** (1): 13–27.
- 10 Hiscock R, Bauld L, Amos A *et al*. Socioeconomic status and smoking: a review. *Ann N Y Acad Sci* 2012; **1248** (1): 107–123.
- 11 O’Keeffe LM, Kearney PM, McCarthy FP *et al*. Prevalence and predictors of alcohol use during pregnancy: findings from international multicentre cohort studies. *BMJ Open* 2015; **5**: e006323.
- 12 Šumilo D, Kurinczuk JJ, Redshaw ME, Gray R. Prevalence and impact of disability in women who had recently given birth in the UK. *BMC Pregnancy Childbirth* 2012; **12** (1): 31.
- 13 Gillman MW, Rifas-Sherman SL, Kleinman K *et al*. Developmental origins of childhood overweight: potential public health impact. *Obesity* 2008; **16**: 1651–1656.
- 14 Robinson SM, Crozier SR, Harvey NC *et al*. Modifiable early-life risk factors for childhood adiposity and overweight: an analysis of their combined impact and potential for prevention. *Am J Clin Nutr* 2015; **101**: 368–375.
- 15 OECD Directorate for Employment Labour and Social Affairs. Obesity update. Paris: OECD, 2014.
- 16 Mallard SR, Connor JL, Houghton LA. Maternal factors associated with heavy periconceptional alcohol intake and drinking following pregnancy recognition: a post-partum survey of New Zealand women. *Drug Alcohol Rev* 2013; **32**: 389–397.
- 17 Morton SMB, Atatoa Carr PE, Grant CC *et al*. Cohort profile: Growing Up in New Zealand. *Int J Epidemiol* 2013; **42** (1): 65–75.
- 18 Morton SMB, Grant CC, Atatoa Carr PE *et al*. How do you recruit and retain a prebirth cohort? Lessons learnt from Growing Up in New Zealand. *Eval Health Prof* 2014; **37**: 411–433.
- 19 Morton SMB, Ramke J, Kinloch J *et al*. Growing Up in New Zealand cohort alignment with all New Zealand births. *Aust N Z J Public Health* 2014; **39** (1): 82–87.
- 20 SNZ: Statistics New Zealand. New Zealand Disability Survey. Wellington: Statistics New Zealand, 2007.
- 21 Huber LRB. Validity of self-reported height and weight in women of reproductive age. *Matern Child Health J* 2007; **11** (2): 137–144.
- 22 Russell D. G., Parnell W. R., Wilson N. C. (1999) NZ Food: NZ People. Key Results of the 1997 National Nutrition Survey. Wellington: Ministry of Health.
- 23 Craig CL, Marshall AL, Sjostrom M *et al*. International physical activity questionnaire: 12-country reliability and validity. *Med Sci Sports Exerc* 2003; **195**: 1381–1395. 195.9131/03 (2003).
- 24 Salmond C, Crampton P, Atkinson J. NZDep2006 Index of Deprivation. Wellington: Department of Public Health, University of Otago, 2007.
- 25 Ministry of Health. Food and Nutrition Guidelines for Healthy Pregnant and Breastfeeding Women: A Background Paper. Wellington: Ministry of Health, 2006.
- 26 NICE: National Institute for Care and Guidance (2010). Quitting Smoking in Pregnancy and Following Childbirth. London: NICE.
- 27 Inskip HM, Crozier SR, Godfrey KM *et al*. Women’s compliance with nutrition and lifestyle recommendations before pregnancy: general population cohort study. *Br Med J* 2009; **338**: 1–6.
- 28 Koren G, Madjunkova S, Maltepe C. The protective effects of nausea and vomiting of pregnancy against adverse fetal outcome—A systematic review. *Reprod Toxicol* 2014; **47**: 77–80.
- 29 Gibbons L, Belizan JM, Lauer JA *et al*. Inequities in the use of caesarean section deliveries in the world. *Am J Obstet Gynecol* 2012; **206**: 331. e1.
- 30 Yerushalmy J. The relationship of parents’ cigarette smoking to outcome of pregnancy—implications as to the problem of inferring causation from observed associations. *Int J Epidemiol* 2014; **43**: 1355–1366.